IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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26-8-2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
  - Data Collection through API
  - Data Collection with Web Scraping
  - Data Wrangling
  - Exploratory Data Analysis with SQL
  - Exploratory Data Analysis with Data Visualization
  - Interactive Visual Analytics with Folium
  - Machine Learning Prediction
- Summary of all results
  - Exploratory Data Analysis result
  - Interactive analytics in screenshots
  - Predictive Analytics result

Introduction

- Project background and context
  - Space X advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against space X for a rocket launch. This goal of the project is to create a machine learning pipeline to predict if the first stage will land successfully.
- Problems you want to find answers
  - What factors determine if the rocket will land successfully?
  - The interaction amongst various features that determine the success rate of a successful landing.
  - What operating conditions needs to be in place to ensure a successful landing program.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Data was collected using SpaceX API and web scraping from Wikipedia.
- Perform data wrangling
  - One-hot encoding was applied to categorical features
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Build, tune, and evaluate classification models to predict first-stage landing success.

Data Collection

- **Data Collection Methodology**
- **SpaceX API** → Launch & rocket data
- **Web Scraping** → Additional launch information
- **Combine Data** → Final dataset for analysis
- **Flow**
- **API + Web Scraping → Data Collection → Dataset**

Data Collection – SpaceX API

- **Key Phrases:**
- GET Request → JSON Response → pd.json_normalize() → Pandas DataFrame

- **GitHub URL**
- **IBM SpaceX Capstone Project** https://github.com/punampatil258/IBM-SpaceX-Capstone-Project

SpaceX REST API
↓
GET Request
↓
JSON Response
↓
JSON Normalization
↓
API Dataset

Data Collection - Scraping

- **Key Phrases:**
- **Wikipedia → BeautifulSoup → HTML Table → Pandas DataFrame**

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Wikipedia Falcon 9 Records
↓
HTML Page
↓
BeautifulSoup
↓
Extract Launch Table
↓
Pandas DataFrame

Data Wrangling

- **Key Activities**
- **Feature Selection** — relevant launch, payload, rocket & landing fields
- **Data Cleaning** — remove/handle incomplete or inconsistent records
- **Data Transformation** — convert data into analysis-ready format
- **One-Hot Encoding** — convert categorical variables into numeric features
- **Final Dataset** — ready for EDA and Machine Learning

- **Data Processing**
- **Raw SpaceX Data**
  → Select Required Columns
  → Clean & Validate Data
  → Handle Missing Values
  → Transform Data
  → Encode Categorical Features
  → **Final Dataset**

GitHub URL

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EDA with Data Visualization

- **Charts & Purpose**
- **Flight Number vs Launch Site** — launch activity by site
- **Payload Mass vs Launch Site** — compare payloads across sites
- **Success Rate vs Orbit Type** — identify landing-success patterns
- **Flight Number vs Orbit Type** — analyze mission evolution
- **Payload Mass vs Orbit Type** — compare payload distribution
- **Yearly Success Trend** — observe landing-success improvement over time

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EDA with SQL

- Summary
  - Load data set to DB2 server
  - Query the names of the unique launch sites in the space mission
  - Query the total payload mass carried by boosters launched by NASA (CRS)
  - Query average payload mass carried by booster version F9 v1.1
  - Query the date when the first successful landing outcome in ground pad was achieved
  - Query the total number of successful and failure mission outcomes
  - etc.
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Build an Interactive Map with Folium

- We marked all launch sites, and added map objects such as markers, circles, lines to
- mark the success or failure of launches for each site on the folium map.
- We assigned the feature launch outcomes (failure or success) to class 0 and 1.i.e., 0 for failure, and 1 for success.
- Using the color-labeled marker clusters, we identified which launch sites have relatively high success rate.
- We calculated the distances between a launch site to its proximities. We answered some question for instance:
    - Are launch sites near railways, highways and coastlines.
    - Do launch sites keep certain distance away from cities.

Build a Dashboard with Plotly Dash

- We built an interactive dashboard with Plotly dash
- We plotted pie charts showing the total launches by a certain sites
- We plotted scatter graph showing the relationship with Outcome and Payload Mass (Kg) for the different booster version.

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Predictive Analysis (Classification)

- **Machine Learning Process**
- **Data Preparation**
  NumPy & Pandas
  Feature Engineering
  Train/Test Split
- **Model Building & Tuning**
  Logistic Regression
  SVM
  Decision Tree
  KNN
  Hyperparameter tuning with **GridSearchCV**
- **Model Evaluation**
  Accuracy metric
  Compare model performance
  Select the **best-performing classifier**

GitHub URL

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Results

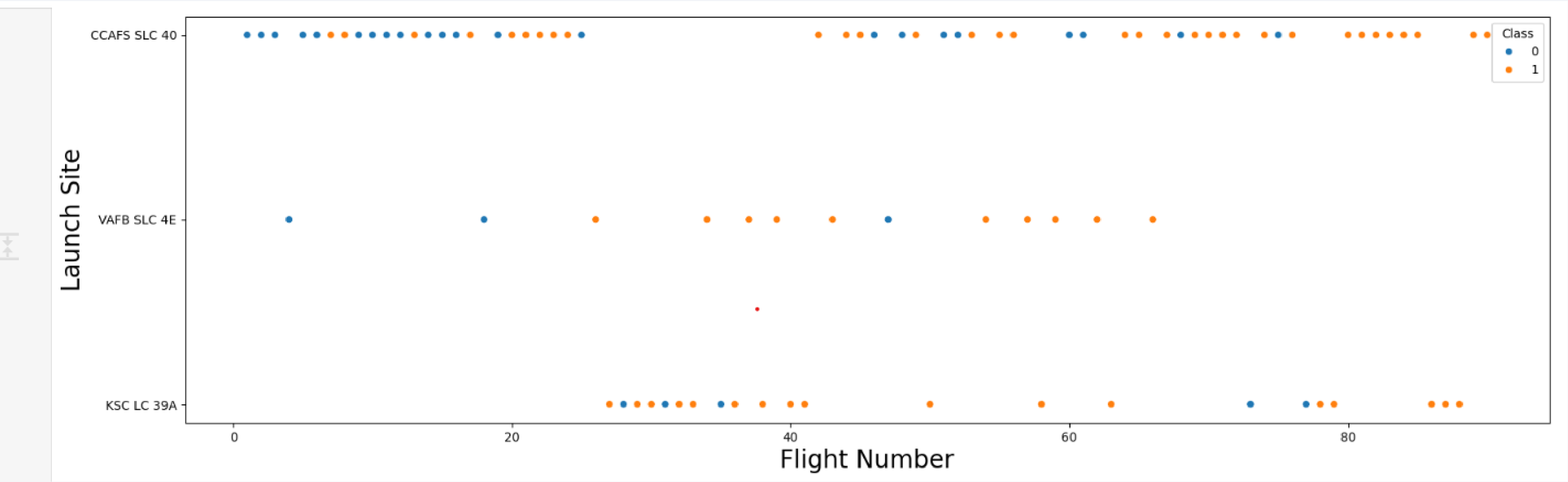
- **Exploratory Data Analysis**
- Visualized trends and relationships between launch attributes.
- Used SQL queries to extract specific information.
- Identified patterns in launch sites, payloads, orbits and landing outcomes.
- **Interactive Analytics**
- **Folium:** Interactive launch-site map.
- **Plotly Dash:** Interactive launch analysis dashboard.
- **Predictive Analysis**
- Classification models were trained and evaluated.
- **Logistic Regression, SVM and KNN: 83.33% test accuracy.**
- **Decision Tree: 77.78% test accuracy.**
- **Highest-performing models: Logistic Regression, SVM & KNN.**

Section 2

Insights drawn from EDA

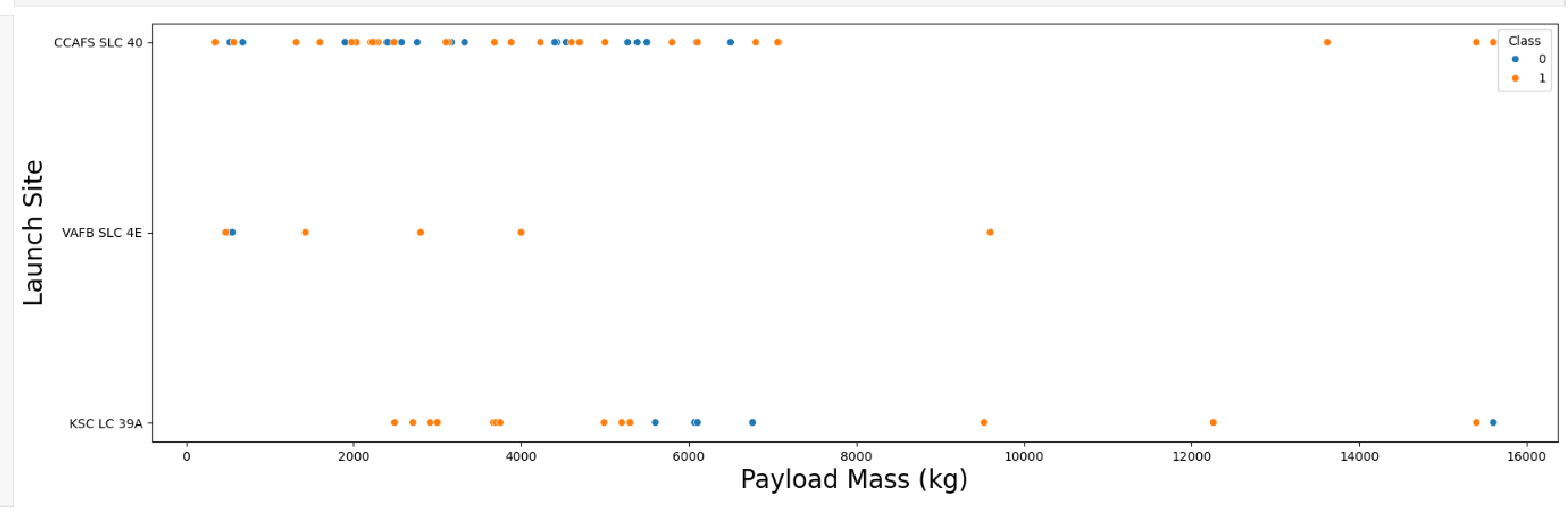
Flight Number vs. Launch Site

The scatter plot shows the distribution of SpaceX flights across different launch sites. Colors represent landing class, helping identify landing-success patterns across launch sites.



Payload vs. Launch Site

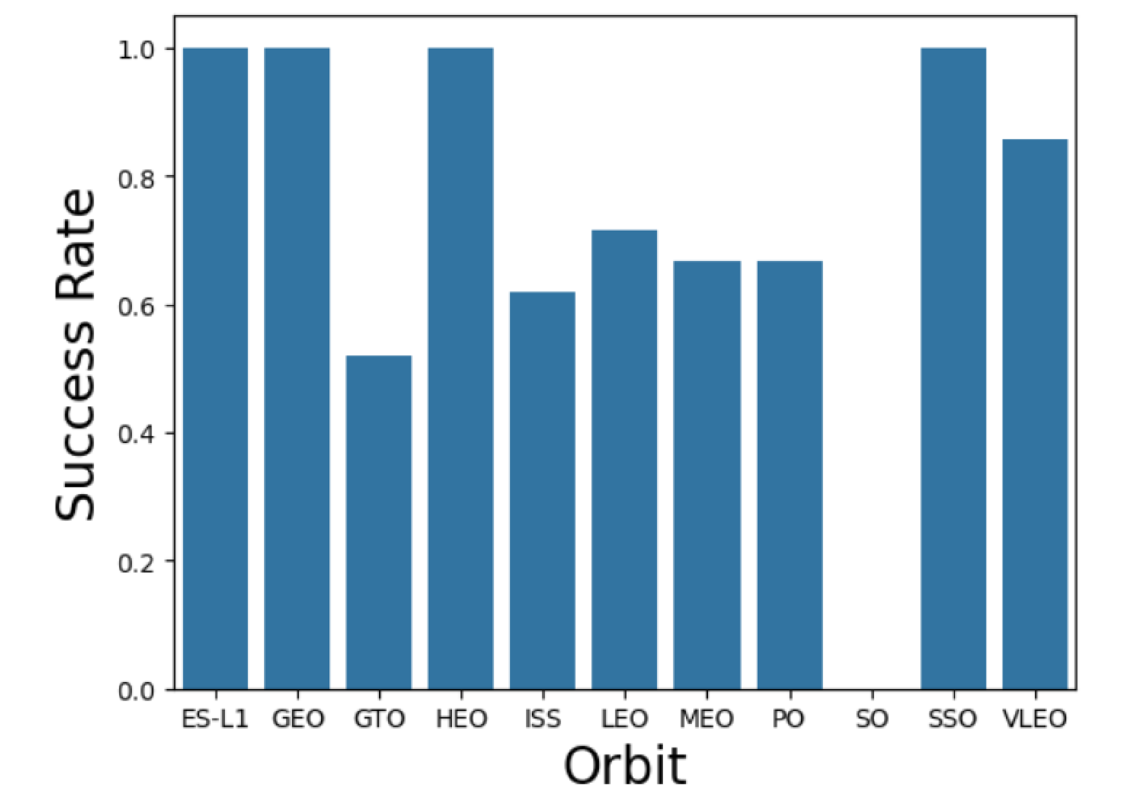
- Scatter plot of Payload vs. Launch Site



- Launch Site VAFB SLC 4E do not have launch that have pay load mass more than 10,000 kg

Success Rate vs. Orbit Type

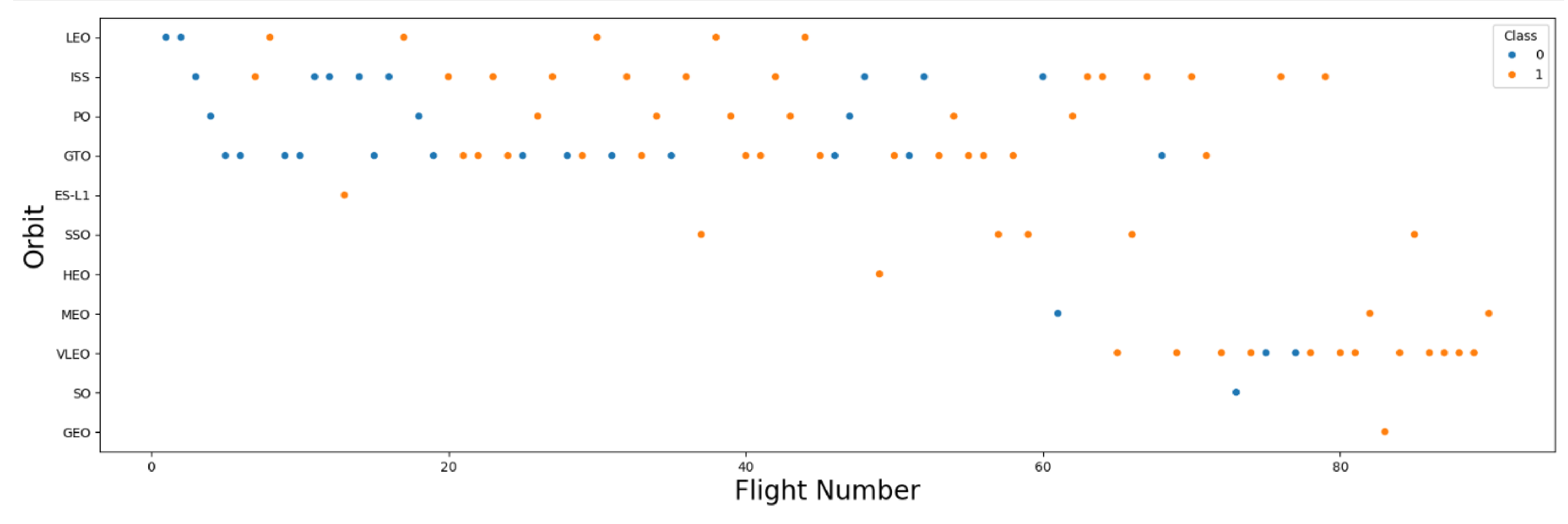
- Bar chart for the success rate of each orbit type



  - Orbit ES-L1, GEO, HEO, SSO have the best success rate

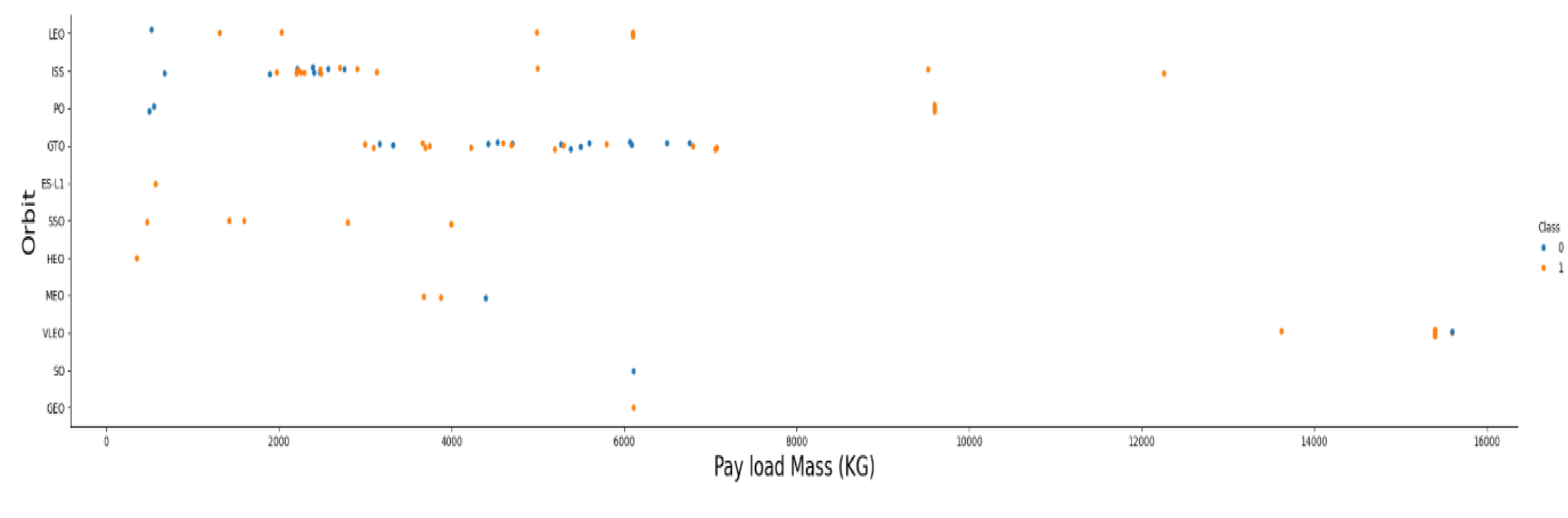
Flight Number vs. Orbit Type

- Scatter point of Flight number vs. Orbit type



Payload vs. Orbit Type

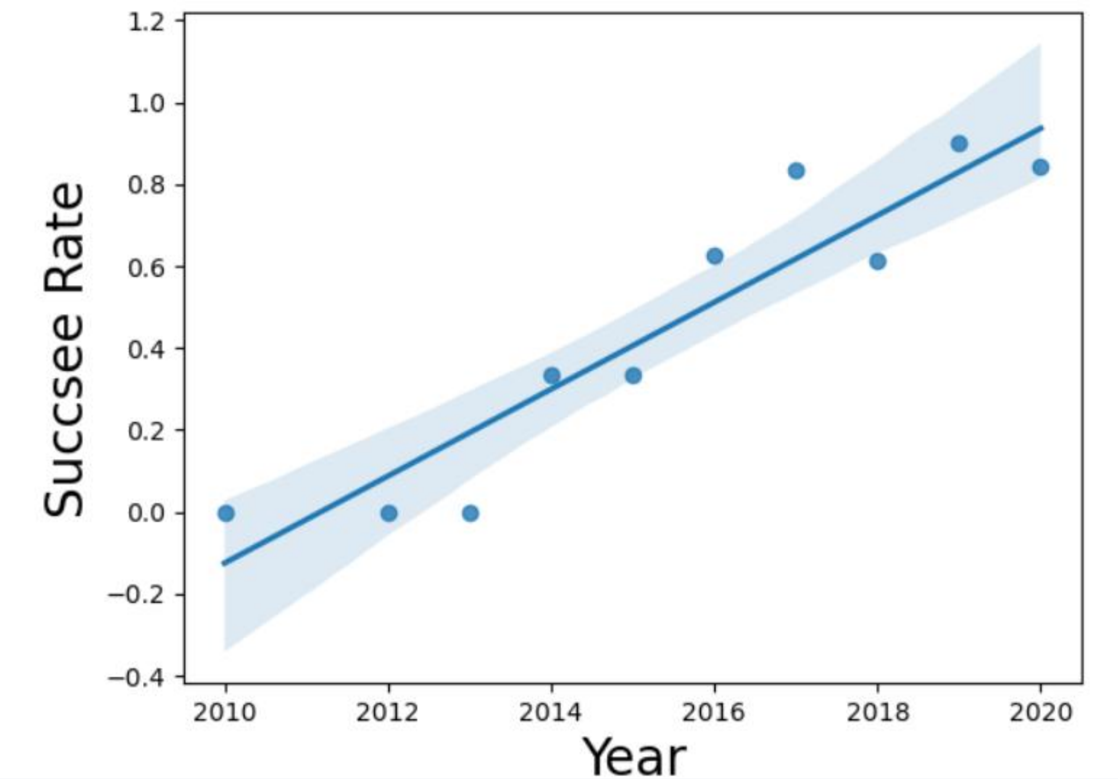
- Scatter point of payload vs. orbit type



- The plot shows the distribution of payload mass across different orbit types. Successful landings are more common for heavier payloads in Polar, LEO, and ISS orbits.

Launch Success Yearly Trend

- Line chart of yearly average success rate



- **The launch success rate shows an improving trend over the years.**

All Launch Site Names

- Find the names of the unique launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- space X have 4 launch sites name above

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- The above are 5 records where launch sites begin with `CCA

Total Payload Mass

- Calculate the total payload carried by boosters from NASA

```
%%sql
SELECT SUM("Payload_Mass__KG_") AS total_payload_mass_kg
FROM SPACEXTABLE
WHERE "Customer" LIKE '%NASA (CRS)%';
```

```
 * sqlite:///my_data1.db
Done.
```

total_payload_mass_kg
48213

- The total payload carried by boosters from NASA is 48213

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

```
%%sql
SELECT AVG("Payload_Mass__KG_") AS average_payload_mass_kg
FROM SPACEXTABLE
WHERE "Booster_Version" = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

average_payload_mass_kg
2928.4

- The average payload mass carried by booster version F9 v1.1 is 2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad

```
%%sql
SELECT MIN("Date") AS first_successful_ground_pad_landing
FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

first_successful_ground_pad_landing
2015-12-22

- The dates of the first successful landing outcome on ground pad is 22 December 2015

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- Total of 4 boosters named above have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes

Landing_Outcome	outcome_count
Controlled (ocean)	5
Failure	3
Failure (drone ship)	5
Failure (parachute)	2
No attempt	21
No attempt	1
Precluded (drone ship)	1
Success	38
Success (drone ship)	14
Success (ground pad)	9
Uncontrolled (ocean)	2

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

- There are 12 total booster which have carried the maximum payload mass

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

- List the failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- There are 2 failed landing outcomes in drone ship in year 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

- There are 8 landing between the date 2010-06-04 and 2017-03-20, where the most common outcome is “No attempt”

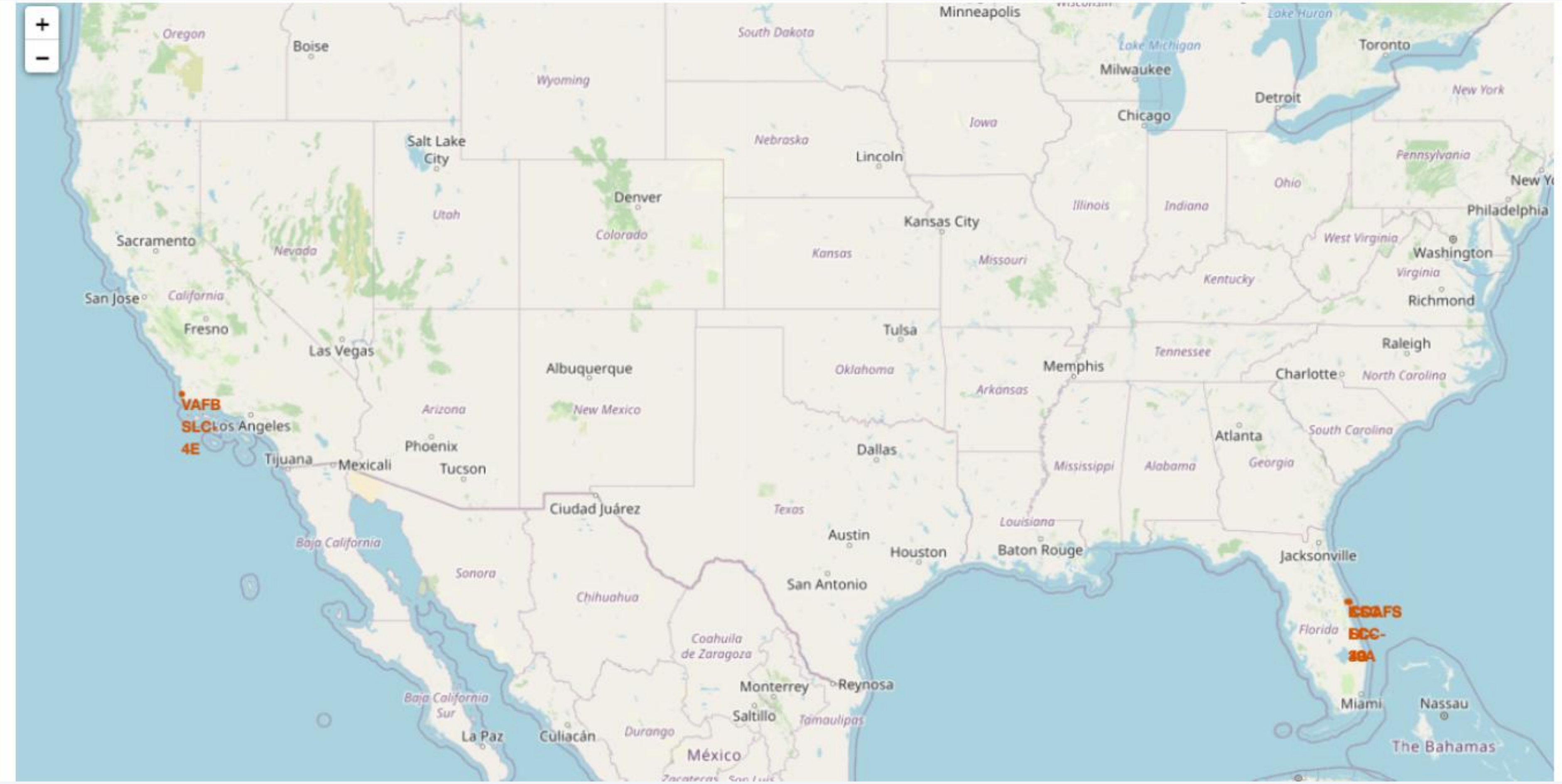
Landing_Outcome	outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

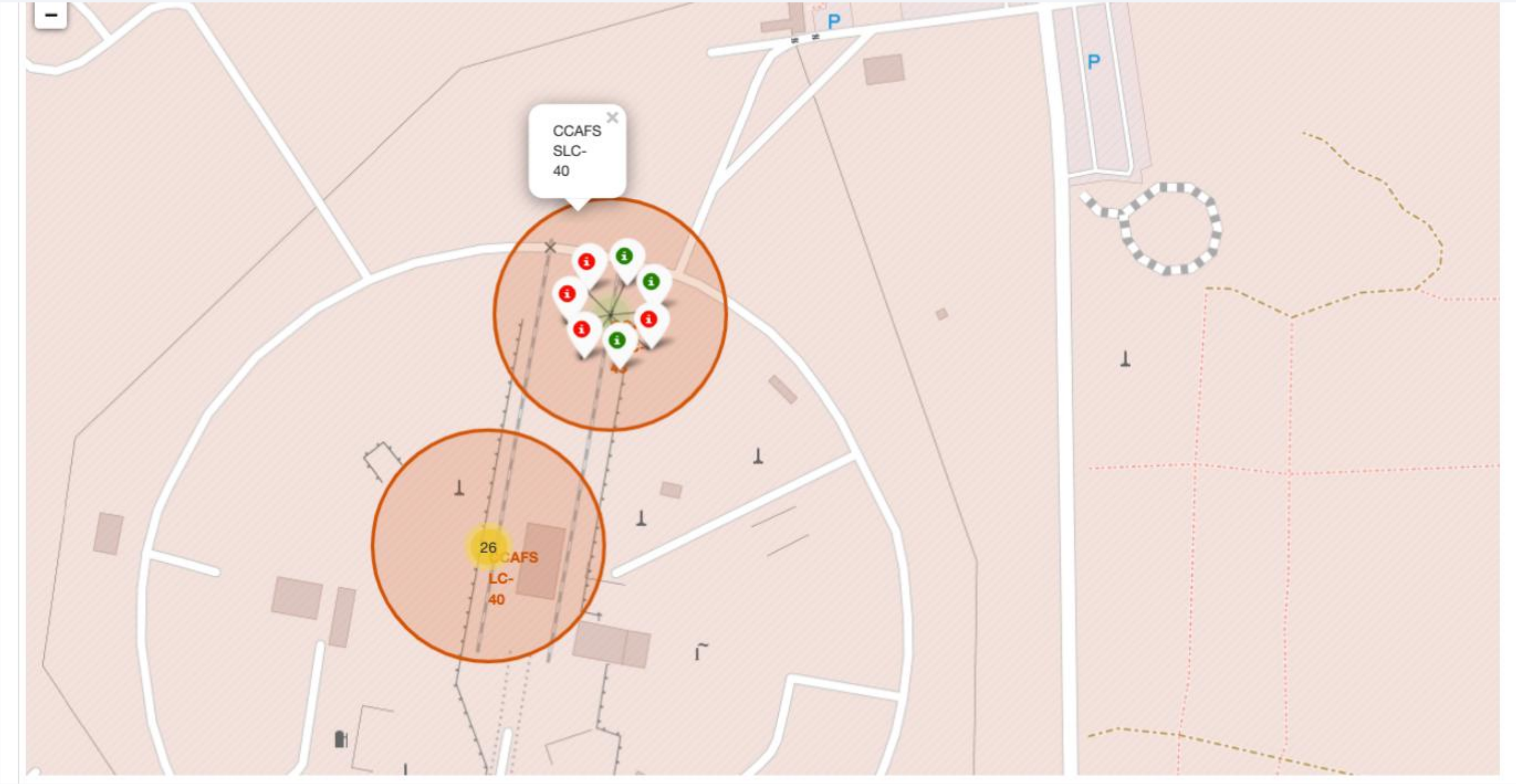
Launch Sites Proximities Analysis

Folium Launch Sites Map

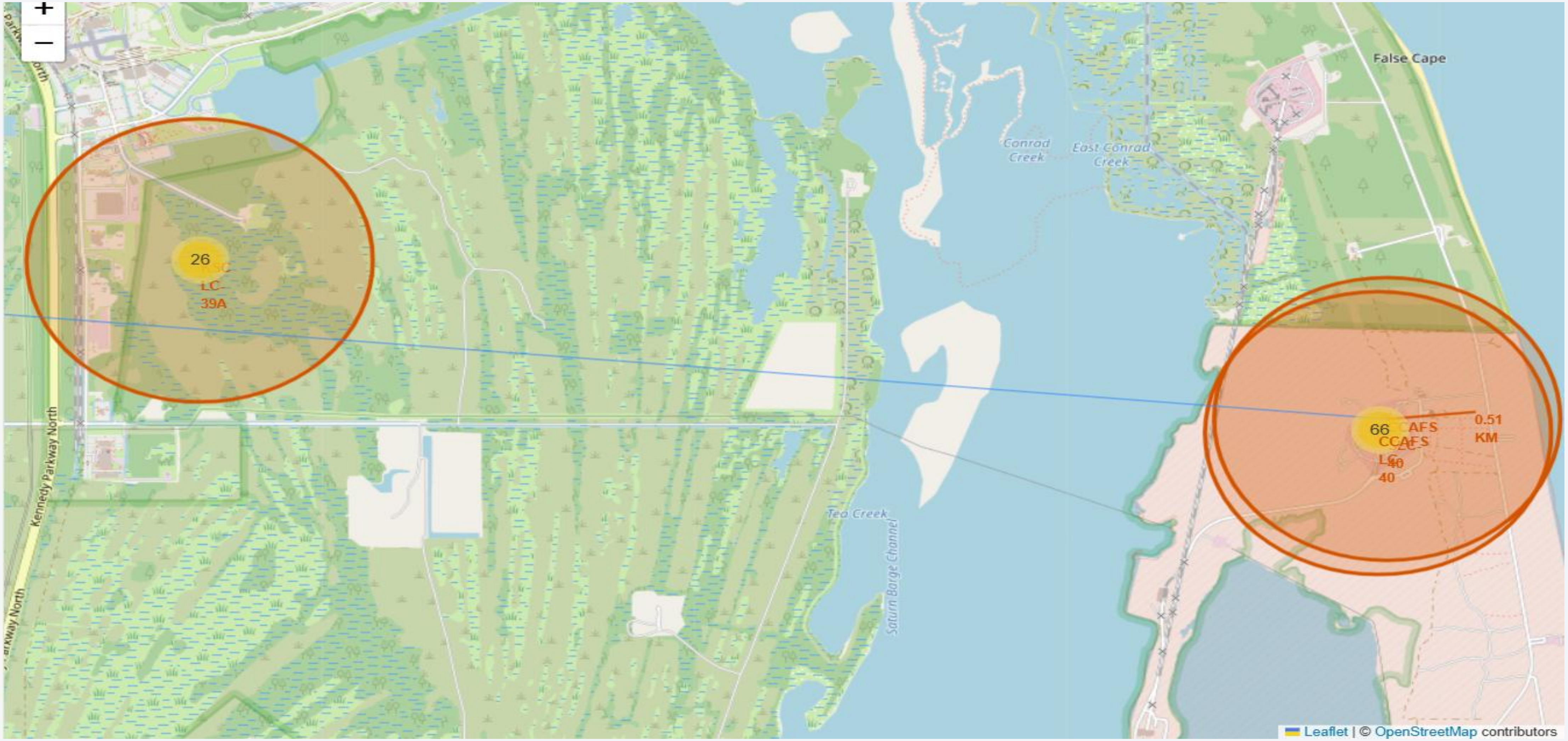
- Create a folium map to mark all launch site's location with circle and marker label



Success & Failed Launches Map Marker



Distances between a Launch Site to its Proximities

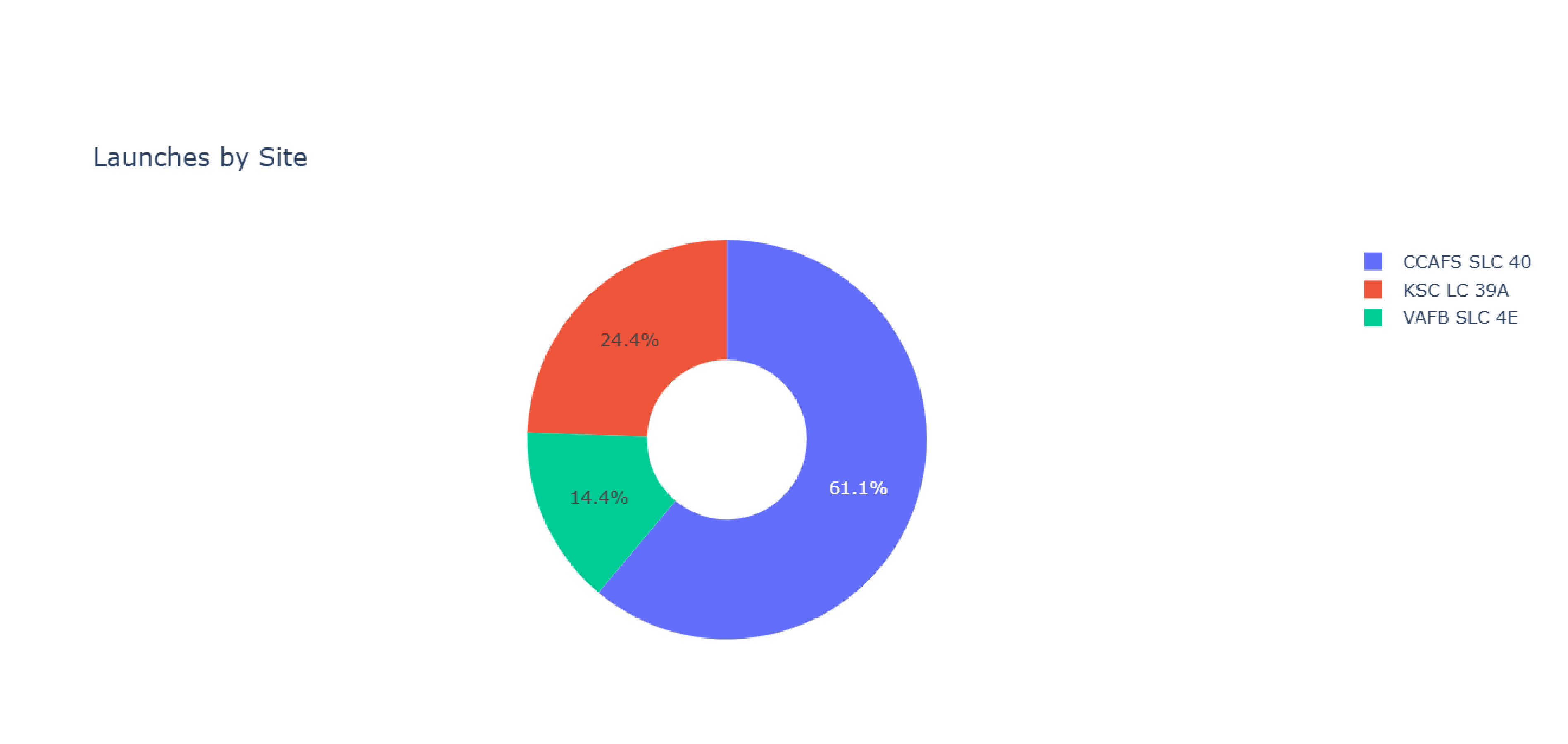


Section 4

Build a Dashboard with Plotly Dash

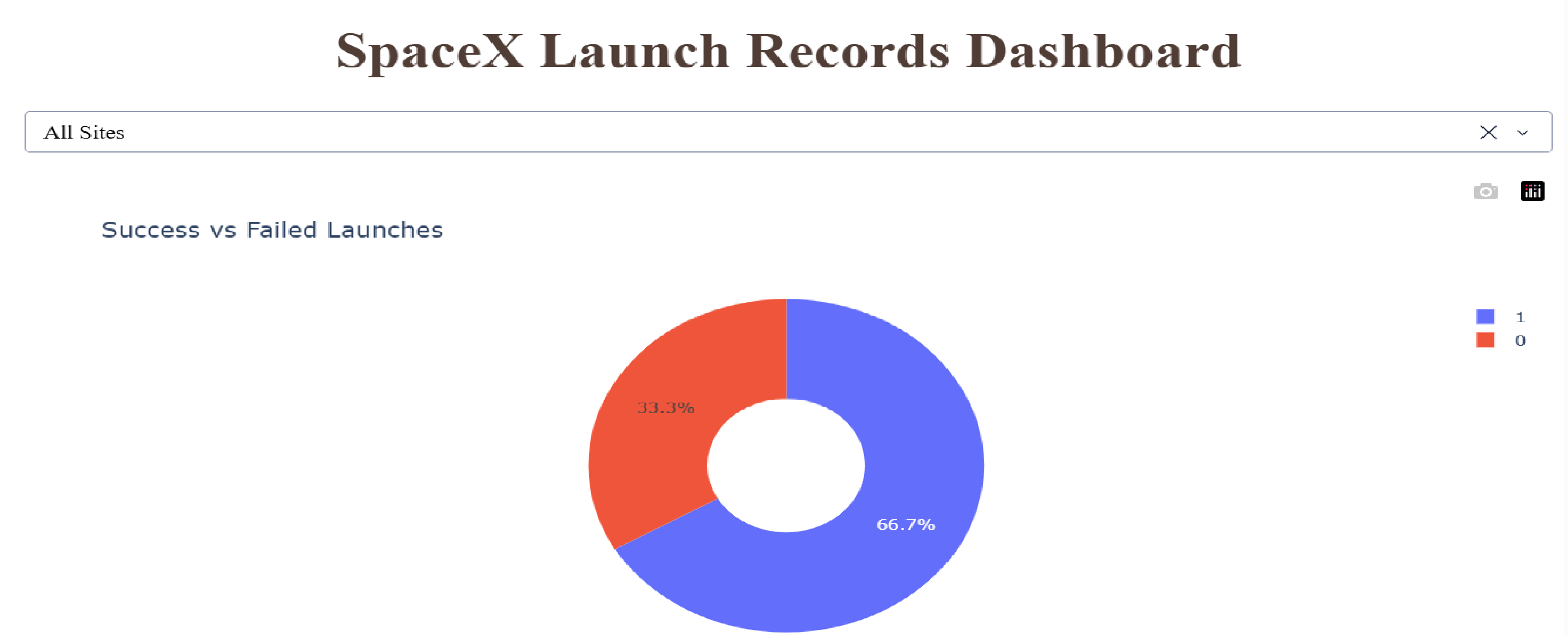
Pie chart showing the success percentage achieved by each launch site

- Dashboard indicate launch success count for all sites in a pie chart



Pie chart showing the Launch site with the highest launch success ratio

- Dashboard indicate the launch site with highest launch success ratio in pie chart



Payload vs. Launch Outcome Dashboard

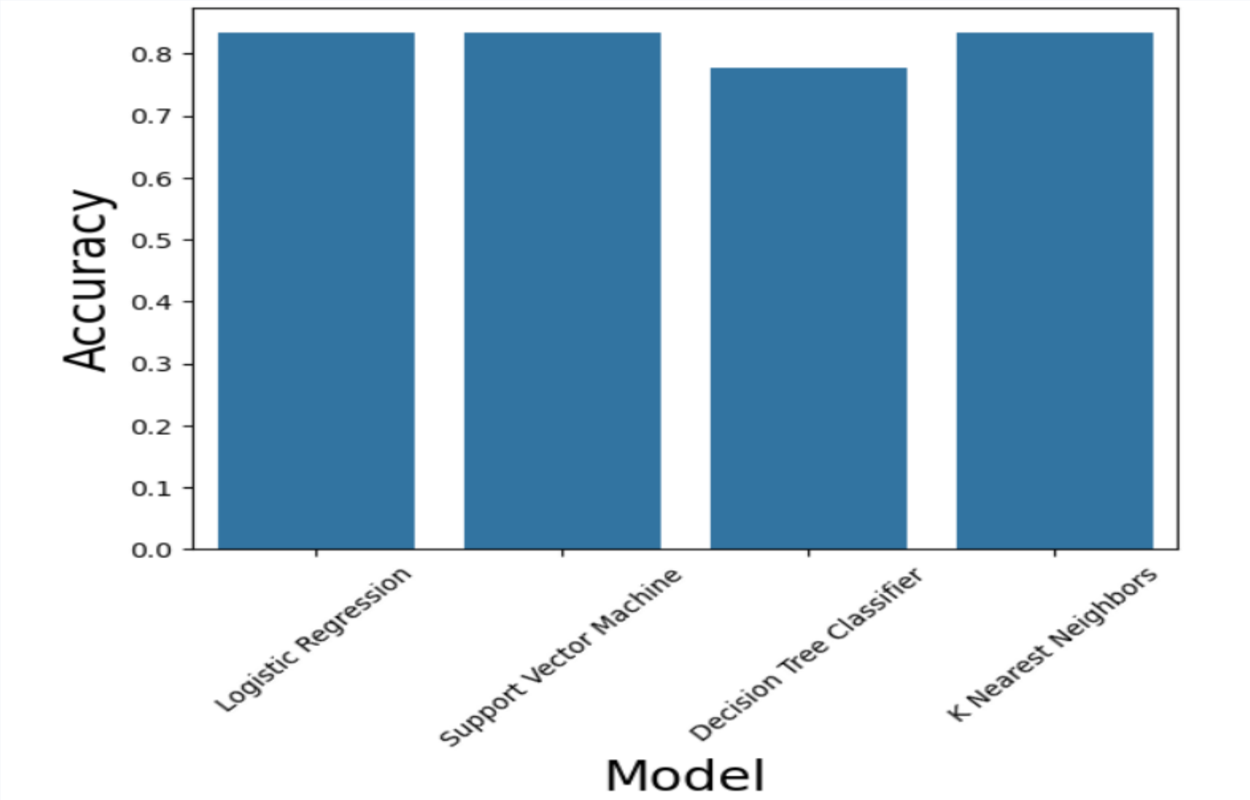
- Dashboard indicate Payload vs. Launch Outcome scatter plot for all sites.

Section 5

Predictive Analysis (Classification)

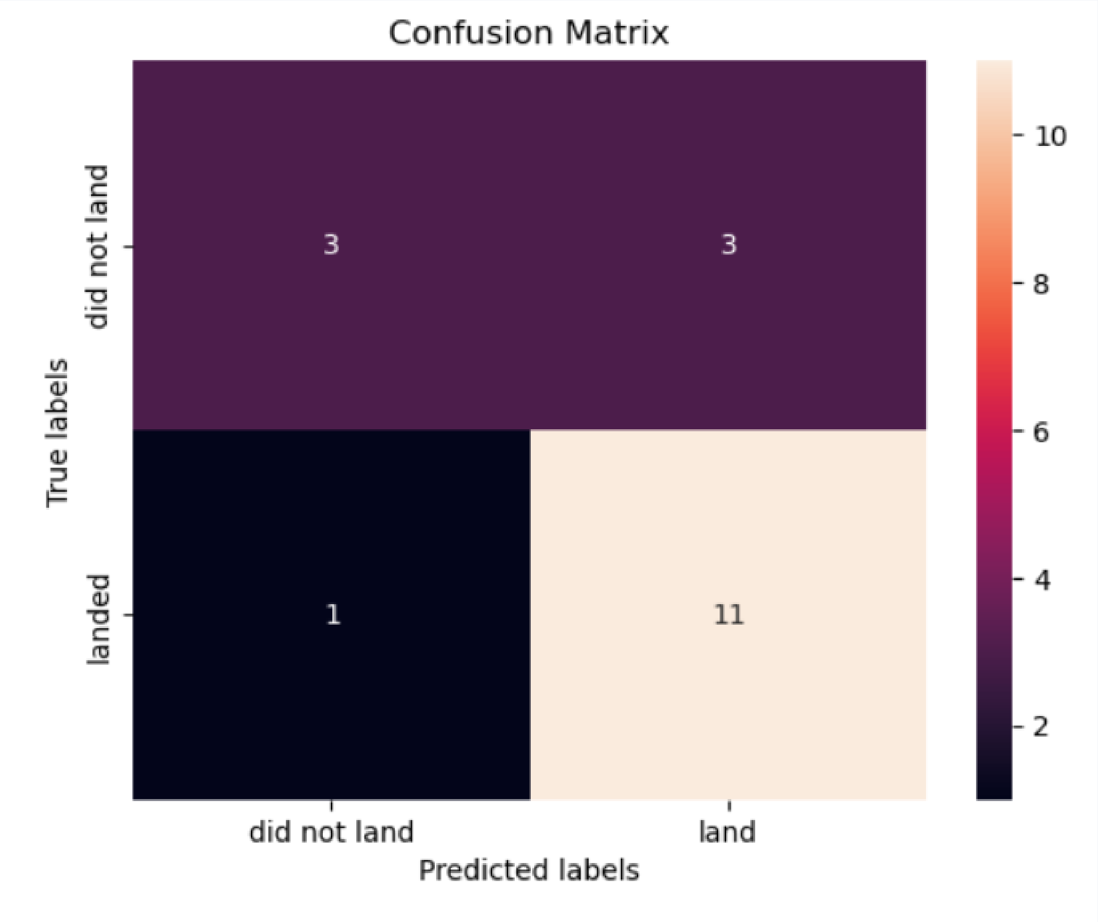
Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart



Confusion Matrix

- Decision Tree Classifier achieved the highest test accuracy. The confusion matrix shows correct predictions on the diagonal and incorrect predictions off the diagonal.



Conclusions

- We can conclude that:
- The larger the flight amount at a launch site, the greater the success rate at a launch site.
- Launch success rate started to increase in 2013 till 2020.
- Orbits ES-L1, GEO, HEO, SSO, VLEO had the most success rate.
- KSC LC-39A had the most successful launches of any sites.
- The Decision tree classifier is the best machine learning algorithm for this task.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project.

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Thank you!